

Fisher Information Calculator

Multi-channel optimal reconstruction of $G(t)$ via Cramér–Rao bound

Model

$Y_i(t) = Y0_i \cdot (1 + c_i \cdot G(t)) + \text{noise}_i$ $\text{noise}_i \sim N(0, \sigma_i^2)$ $G(t) = A_{\text{base}} + A_{\text{amp}} \cdot \cos(2t)$ Fisher Information: $F = \sum_i [Y0_i^2 \cdot c_i^2 / \sigma_i^2]$ Cramér–Rao bound: $\sigma_G^2 \geq 1 / F$ Optimal estimate: $\hat{G}(t) = [\sum_i (c_i \cdot Y0_i / \sigma_i^2)(Y_i(t) - Y0_i)] / F$

Scalar Field Parameters

A_{base}

A_{amp}

Measurement Streams

Add independent measurement channels. Each stream $Y_i(t) = Y0_i \cdot (1 + c_i \cdot G(t)) + \epsilon_i$. More streams $\rightarrow$ tighter Cramér–Rao bound.

Stream	$Y0_i$	c_i	σ_i
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+ Add Stream

Remove

Compute

Results

Fisher Information F

-

Cramér–Rao σ_G^2

-

Cramér–Rao σ_G

-

RMS Recon Error

-

$\hat{G}$ Mean

-

G True Mean

-

$G(t)$ true $\hat{G}(t)$ reconstructed $\pm G$ bound

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